Theoretical Framework

Inductive Coil Launcher Theory

Design Parameters

Quantity	Symbol	Value
Capacitance (each, 2 in series)	C	$0.1\,F\ each \rightarrow C_T = 0.05\,F$
Charge Voltage (capacitor rating)	V_C	$63\,V$
Bank Voltage (series)	V	$126\,V$
Projectile Mass	m	$18.0 \times 10^{-3}\,kg$
Projectile Diameter	d_p	$8.0 \times 10^{-3}\,m$
Projectile Length	l_p	$45.6 \times 10^{-3}\,m$
Projectile Material	$-$	$Soft\ iron\ /\ low-carbon\ steel$
Target Muzzle Velocity	v	$20\,m/s$
Coil Length	l	$45.6 \times 10^{-3}\,m$
Coil Inner Radius	r_i	$5.25 \times 10^{-3}\,m$
Coil Mean Radius	a	$7.44 \times 10^{-3}\,m$
Winding Depth (4 layers)	c	$4.38 \times 10^{-3}\,m$
Wire Diameter	d_w	$1.02 \times 10^{-3}\,m$
Coil Turns	N	164
Coil Resistance	R_{coil}	$0.160\,\Omega$
External Resistance (ESR, switch, leads; estimated)	R_{ext}	$0.060\,\Omega$
Relative Permeability (low-carbon steel)	μ_r	500
Saturation Flux Density (low-carbon steel)	B_{sat}	$1.6\,T$
Permeability of Free Space	μ_0	$4\pi \times 10^{-7}\,H/m$

Total Energy Per Capacitor:

$$E_C = \frac{1}{2}CV^2 = \frac{1}{2}(0.1)(63)^2 = 198.5\,J \approx 200\,J$$

Total Energy in Circuit (Connected in Series)

In series, the voltages of the capacitors add while the total capacitance is reduced:

$$V = V_1 + V_2 = 63 + 63 = 126\,V$$

$$C_T = \frac{C_1 C_2}{C_1 + C_2} = \frac{(0.1)(0.1)}{0.1+0.1} = 0.05\,F$$

$$E_T = \frac{1}{2}C_T V^2 = \frac{1}{2}(0.05)(126)^2 = 397\,J$$

This is the same total energy as the parallel connection, but it is delivered at twice the voltage.

Balancing Resistors:

A 10 kΩ, 1 W resistor is placed across each capacitor so that neither capacitor exceeds its 63 V rating:

$$I_{bal} = \frac{V_C}{R_{bal}} = \frac{63}{10\,000} = 6.3 \times 10^{-3}\,A = 6.3\,mA$$

$$P = \frac{V_C^2}{R_{bal}} = \frac{63^2}{10\,000} = 0.40\,W$$

Desired Kinetic Energy

$$E_K = \frac{1}{2}mv^2 = \frac{1}{2}\left(18.0 \times 10^{-3}\right)\left(20^2\right) = 3.6\,J$$

Efficiency Desired:

$$\eta = \frac{E_K}{E_T} = \frac{3.6}{397} = 9.1 \times 10^{-3} = 0.91\%$$

Projectile Dimensions:

The projectile volume is set by its mass (steel, $\rho = 7850\ kg/m^3$):

$$V_p = \frac{m}{\rho} = \frac{18.0\times10^{-3}}{7850} = 2.29 \times 10^{-6}\,m^3$$

Using a diameter of $d_p = 8.0 \times 10^{-3}\,m$, the length is:

$$l_p = \frac{V_p}{\pi d_p^2/4} = \frac{2.29\times10^{-6}}{5.03\times10^{-5}} = 45.6 \times 10^{-3}\,m$$

$$\lambda = \frac{l_p}{d_p} = \frac{45.6}{8.0} = 5.7$$

Effective Permeability of the Projectile:

A short rod cannot use its full permeability because its own poles oppose the applied field. This is described by the demagnetizing factor N_d, which depends on the aspect ratio λ:

$$N_d = \frac{1}{\lambda^2-1}\left[\frac{\lambda}{\sqrt{\lambda^2-1}}ln\left(\lambda + \sqrt{\lambda^2 - 1}\right) - 1\right] = \frac{1}{5.7^2-1}\left[\frac{5.7}{\sqrt{5.7^2-1}}ln\left(5.7 + \sqrt{5.7^2 - 1}\right) - 1\right] = 0.046$$

$$\mu_{eff} = \frac{\mu_r}{1+N_d(\mu_r-1)} = \frac{500}{1+0.046(499)} = 20.7$$

The projectile saturates once the applied field reaches:

$$B_s = \frac{B_{sat}}{\mu_{eff}} = \frac{1.6}{20.7} = 0.077\ T$$

An 8.0 mm diameter was chosen as the optimum for an 18 g projectile. Thicker rods lose effective permeability to demagnetization, while thinner rods reach the saturation limit and need a longer coil, which reduces the work done.

Projectile Diameter	Length / Aspect Ratio	Relative Work
5 mm	117 mm / 23	0.33
7 mm	60 mm / 8.5	0.78
8 mm	46 mm / 5.7	1.00
10 mm	29 mm / 2.9	0.67
12 mm	20 mm / 1.7	0.50

Coil Inductance (Short Multilayer Coil)

Coil Resistance:

$$R = R_{coil} + R_{ext} = 0.160 + 0.060 = 0.220\ \Omega$$

As $l/a \approx 6$ and the coil has 4 layers, the long-solenoid approximation is no longer valid. Wheeler's multilayer formula is used instead (a, l, c in metres, L in μH):

$$L = \frac{31.5\ a^2N^2}{6a+9l+10c} = \frac{31.5\left(7.44\times10^{-3}\right)^2(164)^2}{6\left(7.44\times10^{-3}\right)+9\left(45.6\times10^{-3}\right)+10\left(4.38\times10^{-3}\right)} = \frac{46.9}{0.499} = 94\ \mu H$$

Circuit Behaviour (Series RLC Discharge)

The capacitor discharging through the coil is a series Resistor, Inductor, and Capacitor (RLC) circuit. Its behaviour depends on the damping:

$$R_{critical} = 2\sqrt{\frac{L}{C_T}} = 2\sqrt{\frac{94\times10^{-6}}{0.05}} = 0.087\ \Omega$$

Since $R = 0.220\ \Omega > R_{critical}$, the circuit is overdamped; therefore, the current would rise and then decay without oscillating. This can be described in two-time constants:

Current Rise (Inductive):

$$\tau_L = \frac{L}{R} = \frac{94\times10^{-6}}{0.220} = 4.27\times10^{-4}\ s = 0.43\ ms$$

Current Decay (Capacitive):

$$\tau_C = RC_T = (0.220)(0.05) = 0.011\,s = 11\,ms$$

Peak Current:

The resistive limit of the current is:

$$I_{max} = \frac{V}{R} = \frac{126\,V}{0.220\,\Omega} = 573\,A$$

As τ_L is no longer negligible compared to τ_C, the capacitor voltage drops while the current is still rising, so the simulated peak current is lower:

$$I_{peak} \approx 519\,A$$

Magnetic Field at Coil Centre:

$$B = \mu_0 \frac{NI}{l} = \left(4\pi \times 10^{-7}\right)\frac{(164)(519)}{45.6\times10^{-3}} = 2.35\,T$$

Summing the field of each loop of the finite multilayer coil gives $B_{centre} \approx 2.26\,T$, and an average over the projectile of $3.86 \times 10^{-3}\,T/A$. Both are far above $B_s = 0.077\,T$, so the projectile is fully saturated.

Work and Force on the Projectile:

A ferromagnetic projectile is pulled toward the region of highest inductance (the coil center). The force follows from the change in inductance with the projectile position x:

$$F = \frac{1}{2} I^2 \frac{dL}{dx}$$

Taking the integral from entry to the coil centre (at roughly constant current), the work done on the projectile would be:

$W = \frac{1}{2} I^2\,\Delta L$, given that $\Delta L = L_{with\ slug} - L_{air}$

This replaces the original "Number of Coil's" Step in my previous calculations which set the coil's stored energy as $\frac{1}{2} L I^2$ equal to E_k. That isn't the energy-transfer mechanism needed, however, as the capacitor keeps supplying current while the slug moves, and the work on the slug will depend on how much the slug changes the inductance.

Since the projectile is saturated, its magnetization is capped at M_s, so the work from entry to the coil centre becomes:

$$M_s = \frac{B_{sat}}{\mu_0} = \frac{1.6}{4\pi\times10^{-7}} = 1.27 \times 10^6\ A/m$$

$$W \approx V_p M_s\left(B^- - \frac{1}{2}B_s\right)$$

Using an average current over the stroke of about 450 A:

$$B^- = \left(3.86 \times 10^{-3}\right)(450) = 1.74\ T$$

$$W = \left(2.29 \times 10^{-6}\right)\left(1.27 \times 10^6\right)(1.74 - 0.039) = 4.96\ J$$

Muzzle Velocity:

$$v = \sqrt{\frac{2W}{m}} = \sqrt{\frac{2(4.96)}{18.0\times10^{-3}}} = 23.5\ m/s$$

A time-domain simulation of the circuit and projectile together gives 23.1 m/s (4.78 J). The model does not include eddy currents, friction, or back-EMF, which typically cost 15–30% of the kinetic energy, so the expected muzzle velocity is about 18–21 m/s.

Number of Turns Required on the Coil:

The number of turns is set by the coil length and the wire diameter (1.02 mm wire, about 1.10 mm with enamel):

$$N_{layer} = \frac{l}{d_w} = \frac{45.6\times10^{-3}}{1.10\times10^{-3}} = 41\ turns$$

$$N = (layers)\left(N_{layer}\right) = (4)(41) = 164\ turns$$

The wire length and coil resistance are (1.02 mm copper wire: $0.0209\ \Omega/m$):

$$l_{wire} = \sum_{j=1}^{4} 2\pi r_j N_{layer} = 7.67\ m$$

$$R_{coil} = (7.67)(0.0209) = 0.160\ \Omega$$

Four layers were chosen as the balance between ampere-turns (NI) and peak current. Fewer layers raise the current, which needs a larger switch, while more layers add resistance with little gain in speed:

Layers (Turns)	Peak Current	Simulated Velocity
2 (82)	919 A	19.4 m/s
3 (123)	675 A	22.0 m/s
4 (164)	519 A	23.1 m/s
5 (205)	412 A	23.4 m/s

Current Cut-off Timing:

As the coil length matches the projectile length, the projectile's centre reaches the coil centre when its nose reaches the coil exit. An optical sensor at the coil exit turns the switch off at this point (simulated at $t_{cut} \approx 4.35\,ms$). Without the cut-off, $\tau_C = 11\,ms$ is longer than the stroke, and the projectile is pulled back to about 6.1 m/s.

The energy left in the coil at cut-off is absorbed by a flyback diode across the coil:

$$E_L = \frac{1}{2} L I_{cut}^2 = \frac{1}{2}\left(94 \times 10^{-6}\right)(412)^2 = 8.0\,J$$

Efficiency Achieved:

After the cut-off, the capacitor bank is left at $V_{end} = 87.2\,V$:

$$E_{drawn} = \frac{1}{2} C_T \left(V^2 - V_{end}^2\right) = \frac{1}{2}(0.05)\left(126^2 - 87.2^2\right) = 207\,J$$

$$\eta_{drawn} = \frac{4.78}{207} = 2.3\%$$

$$\eta_{total} = \frac{4.78}{397} = 1.2\%$$

Heat Per Shot:

$$m_{Cu} = \rho_{Cu} A_{wire} l_{wire} = (8960)\left(0.823 \times 10^{-6}\right)(7.67) = 56.6 \times 10^{-3}\,kg$$

$$\Delta T = \frac{E_{drawn} - E_K}{m_{Cu} c_{Cu}} = \frac{207 - 4.8}{\left(56.6 \times 10^{-3}\right)(385)} = 9.3\,°C$$

Component Ratings:

Component	Rating	Notes
Switch (IGBT / MOSFET)	≥ 200 V, ≥ 600 A pulse	Must be able to turn off; an SCR cannot
Flyback Diode	≥ 200 V, ≥ 450 A pulse	Fast-recovery, across the coil
Balancing Resistors	10 kΩ, 1 W	One across each capacitor
Charger	126 V output	Stops automatically at 126 V
Wiring	≥ 4 mm², insulation > 126 V	Keep leads short (counts toward R_ext)

Figure 1: Rough Design Schematics

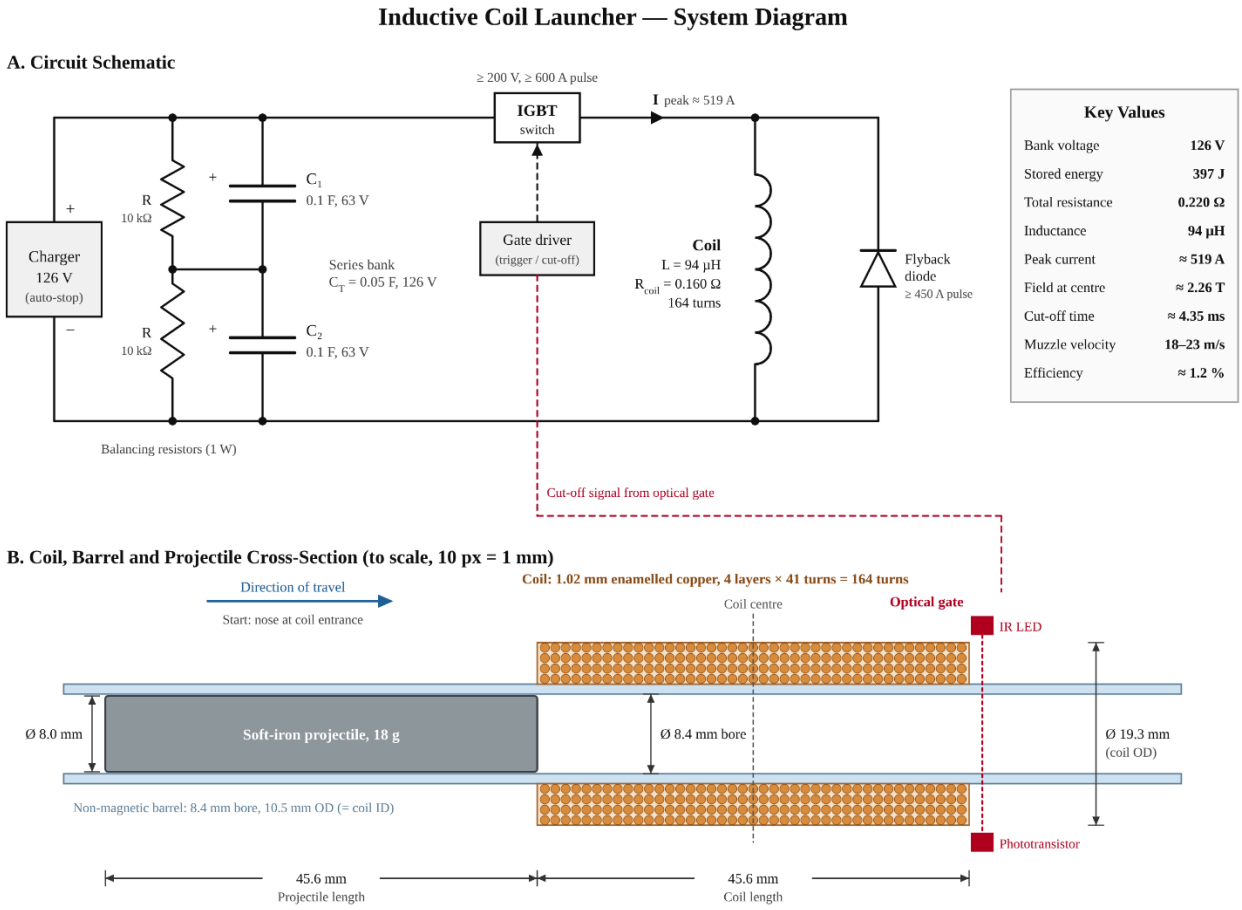
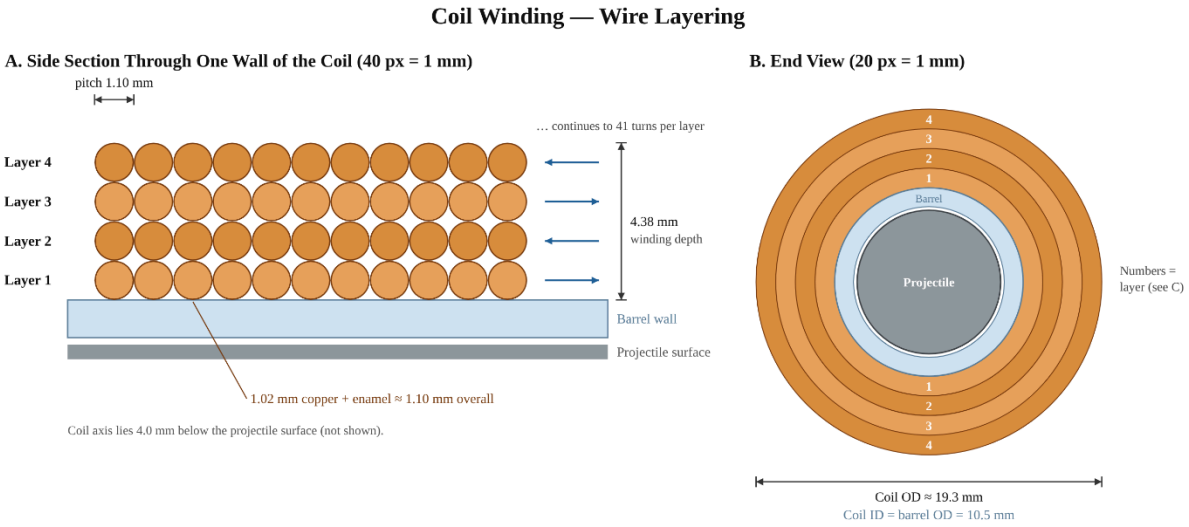


Figure 2: Coil Wiring Diagram



C. Layer Radii and Wire Length (41 turns per layer)

Layer	Mean radius, r_j	Circumference, $2\pi r_j$	Wire per layer, $2\pi r_j \times 41$	Axial winding direction
1	5.80 mm	36.4 mm	1.49 m	Breech → muzzle
2	6.89 mm	43.3 mm	1.78 m	Muzzle → breech
3	7.99 mm	50.2 mm	2.06 m	Breech → muzzle
4	9.08 mm	57.1 mm	2.34 m	Muzzle → breech
Total			**7.67 m → R = 7.67 × 0.0209 = 0.160 Ω**	

Winding note: wind every layer in the same rotational direction (e.g. clockwise viewed from the breech); only the axial direction reverses at the end of each layer. With an even number of layers, both leads finish at the breech end of the coil.